

MANIPAL UNIVERSITY JAIPUR

Department of Department of Artificial Intelligence & Machine Learning

Course Hand-out

A. Basic Details:

Programme Name:	MTECH (AIML)
Course Name:	APPLIED STATISTICS AND PROBABILITY
Course Code:	AIM6101
LTPC (<i>Lecture Tutorial Practical Credits</i>):	3 – 1 – 0 - 4
Session:	AUG - DEC 25
Class:	I
Course Coordinator:	Dr. Rahul Sharma
Course Instructor(s):	Dr. Rahul Sharma

B. Introduction: This course, offered by the Department of AIML, is designed for M.Tech students to build a strong foundation in statistical methods essential for research and advanced applications in Computer Science, IT, and Communication Engineering. It covers the role of statistics in engineering through basic principles, types of studies (retrospective, observational, and designed experiments), and process observation. Students will learn measures of central tendency (mean, median, mode), dispersion (range, quartile deviation, mean and standard deviation), skewness, and kurtosis. The course introduces probability concepts, including sample spaces, probability rules, Bayes' theorem, and distributions such as Bernoulli, Binomial, Poisson, and Normal. It also covers correlation and regression using Karl Pearson, Spearman, and least squares methods. Advanced topics include discrete and continuous random variables, joint distributions, and hypothesis testing. The course equips students with analytical tools to interpret data and support algorithm development and research.

COURSE OBJECTIVES

This course aims to introduce fundamental statistical concepts and techniques essential for analyzing data and making informed decisions in engineering and research contexts. The course is designed to provide students with both practical skills and theoretical understanding of statistical tools, emphasizing their application in real-world engineering problems.

- Explain the role of statistics in engineering and apply basic statistical principles.
- Compute and interpret measures of central tendency, dispersion, skewness, and

kurtosis.

- Apply probability theory and probability distributions to model uncertainty in engineering problems.
- Analyse relationships between variables using correlation and regression techniques.
- Formulate and test hypotheses using appropriate statistical methods for informed decision making.

C. Course Outcomes: At the end of the course, students will be able to:

<i>CO Statement</i>	<i>CO</i>	<i>Bloom's Level</i>	<i>Target Attainment %</i>	<i>Target Attainment level</i>
AI6101.1	Explain the foundational principles of statistics and their applications in engineering contexts.	Remember and/or understand	80	3
AI6101.2	Apply appropriate measures of central tendency, dispersion, skewness, and kurtosis to analyse data sets.	Understand and/or apply	80	3
AI6101.3	Analyse various probability distributions use them to model random processes.	Analyse and/or create	80	3
AI6101.4	Evaluate relationships between variables using correlation and regression techniques.	Analyse and/or evaluate	80	3
AI6101.5	Perform hypothesis testing using appropriate statistical tests to draw inferences from data.	Evaluate and Create/ Apply	80	3

D. Program Outcomes and Program Specific Outcomes:

At the end of the MTech AIML program, the student will be able:

PROGRAM OUTCOMES (PO's)

[PO.1] An ability to independently carry out research /investigation and development work to solve practical problems.

[PO.2] An ability to write and present a substantial technical report/document.

[PO.3] Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.

PROGRAM SPECIFIC OUTCOMES (PSO's)

[PSO.1] Students will be able to examine the applications of Artificial Intelligence and Machine Learning in real-life problems.

[PSO.2] Students will be able to design and implement intelligent systems for multi-disciplinary problems.

E. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Formative and Summative)	Mid-Term Examination (Close) [60% of the syllabus]	30
	Class Work Sessional (CWS): Quiz, MOOC, Research work Assignment, Video Assignment, Written Assignment	30
End Term Exam (Summative)	End Term Exam (Close)	40
	Total	100

F. Syllabus:

Basics of Statistics: The Role of Statistics in Engineering, Basic Principles, Retrospective Study, Observational study, Designed Experiments, Observing Processes Over Time, Mechanistic and Empirical Models, Probability and Probability Models.

Measure of central tendency: mean, median and mode, Measures of dispersion- Range, Quartile Deviation, Mean Deviation, Standard Deviation, Coefficient of variance, Skewness, Kurtosis.

Probability Distribution: Sample Spaces and Events, Interpretations and Axioms of Probability, Addition Rules, Conditional Probability, Multiplication and Total, Probability Rules, Bayes' Theorem, Random Variables, Concept of Random Variable, Bernoulli Distribution, Binomial Distribution, Poisson Distribution, Normal Distribution.

Correlation and Regression: Concept and types, Karl Pearson Method, Rank Spearman Method, Least Square Method, Discrete Random Variables and Probability Distributions. Continuous Random Variables and Probability Distributions. Joint Probability Distributions.

Testing of Hypothesis: Testing of Hypothesis, Null and alternative hypothesis, level of significance, one-tailed and two-tailed tests, tests for large samples (tests for single mean, difference of means, single proportion, difference of proportions), t-test, F- test, Chi-Square Test.

References:

- Douglas C. Montgomery and George C. Runger, Applied Statistics and Probability for Engineers, John Wiley & Sons.
- Blake I., An Introduction to Applied Probability, John Wiley & Sons.
- Yagolam A. M. and Yagolam I. M., Probability and Information, Hindustan Publishing Corporation, Delhi, 1983.

G. Lecture Plan:

Lecture No.	Topic	Session Outcome	Corresponding CO	Mode of delivery	Mode of assessing CO
1	Course outcome and syllabus discussion.	Understanding the course structure of applied statistics and probability.	AIM6101.1	Discussion	NA
2	Basics of statistics	To learn basics of statics	AIM6101.1	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
3	The role of statistics in engineering	To learn basics of statics and their roles in engineering.	AIM6101.1	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
4	Basic principles, retrospective study, observational study	Understands statics principles and study related observation.	AIM6101.1	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
5	Designed experiments	To understands track and trends.	AIM6101.1	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
6	Designed experiments, observing processes over time	To understands track trends and causes through experimentation.	AIM6101.1	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE

7	Mechanistic and empirical models.	Learn Model systems through logic and observation.	AIM6101.1	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
8	Probability and probability models	Understand chance with probability-based models.	AIM6101.1	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
9	Measure of central tendency: mean, median and mode	Understand and calculate mean, median, and mode.	AIM6101.2	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
10	Measures of dispersion Range	Understand and calculate range and quartile deviation.	AIM6101.2	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
11	Measures of Quartile Deviation	Understand and calculate range and quartile deviation.	AIM6101.2	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
12	Mean Deviation, Standard Deviation	Understand and calculate mean and standard deviation.	AIM6101.2	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
13	Coefficient of variance, Skewness, Kurtosis.	Understand and interpret coefficient of variance, skewness, and kurtosis.	AIM6101.2	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE

14	Probability Distribution	Understand sample spaces and types of events.	AIM6101.3	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
15	Sample Spaces and Events.	Understand sample spaces and types of events.	AIM6101.3	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
16	Interpretations and Axioms of Probability	To understand and apply axioms of probability.	AIM6101.3	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
17	Addition Rules, Conditional Probability	Understand addition rules and conditional probability.	AIM6101.3	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
18	Multiplication and Total	Multiplication and total rules determine outcome probabilities.	AIM6101.3	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
19	Probability Rules	Multiplication and total rules determine outcome probabilities.	AIM6101.3	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
20	Bayes' Theorem, Random Variables	Bayes' Theorem updates probabilities: random variables vary outcomes.	AIM6101.3	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE

21	Concept of Random Variable, Bernoulli Distribution	Random variables represent outcomes; Bernoulli models binary events.	AIM6101.3	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
22	Binomial Distribution	To understand Binomial counts successes.	AIM6101.3	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
23	Poisson Distribution, Normal Distribution.	To understand Poisson rare events; Normal continuous data.	AIM6101.3	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
Mid Term Examination					
24	Correlation and Regression: Concept and types	Understand concepts and types of correlation and regression.	AIM6101.4	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
25	Karl Pearson Method Rank Spearman Method	Discuss Karl Pearson measures linear	AIM6101.4	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
26	Rank Spearman Method	Correlation Spearman Rank measures correlation of ranked data.	AIM6101.4	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
27	Least Square Method	Understand least squares for best-fit regression.	AIM6101.4	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE

28	Discrete Random Variables	Discrete random variables have countable outcomes.	AIM6101.4	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
29	Probability Distributions	Probability distributions assign their probabilities.	AIM6101.4	Lecture	Class Test Quiz, Seminar, Assignment, MTE/ETE
30	Continuous Random Variables	Understand continuous variables and joint probabilities.	AIM6101.4	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
31	Probability Distributions. Joint Probability	Understand continuous variables and joint probabilities.	AIM6101.4	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
32	Distributions. Testing of Hypothesis	Master hypothesis testing and decision making.	AIM6101.4	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
33	Testing Hypothesis of	Master hypothesis testing and decision making.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
34	Null and alternative hypothesis	Define and distinguish null versus alternative hypotheses.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE

35	Level of significance.	Explain significance levels and tail test types.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
36	one-tailed and two-tailed tests	Explain significance levels and tail test types.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
37	Tests for the large samples (tests for single mean, difference of means)	Perform large sample tests for means and proportions.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
38	Tests for large samples (tests for single proportion difference of proportions)	Perform large sample tests for means and proportions.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
39	T-test, F- test	Apply t-test and F-test for analysis.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
40	Chi-Square Test	Use Chi-Square test for categorical data.	AIM6101.5	Lecture Demonstration	Class Test Quiz, Seminar, Assignment, MTE/ETE
41	Research Paper - 1 Discussion related to numerical based	To understand related testing and results.	AIM6101.1	Case Study	Group- teaching and learning
42	Research Paper - 2 Discussion related to numerical based	To understand related testing and results.	AIM6101.2	Case Study	Group- teaching and learning

43	Research Paper - 3 Discussion related to numerical based	To understand related testing and results.	AIM6101.3	Case Study	Group- teaching and learning
44	Revision	Discuss chapter revision and advanced topic and practical techniques	AIM6101.1	NA	Group- teaching and learning
45	Revision	Discuss chapter revision and advanced topic and practical techniques	AIM6101.2	NA	Learning through problem-solving
46	Revision	Discuss chapter revision and advanced topic and practical techniques	AIM6101.3	NA	Learning through problem-solving
47	Revision	Discuss chapter revision and advanced topic and practical techniques	AIM6101.4	NA	Learning through problem-solving
48	Revision	Discuss chapter revision and advanced topic and practical techniques	AIM6101.5	NA	Learning through problem-solving
End Term Examination					

Outside class engagement activities (To complete the 30 Notional hours for each credit as per NCrf guidelines):

Type of Activity	Difficulty Level	Hours Engaged
Statistical Analysis on Real Dataset	Medium–High	10 hours
Research Paper Review: Read and summarize 2 recent papers using statistics in engineering or AI/ML	Medium	5 hours
Case Study Analysis: Solve case studies using probability and statistics in real applications	Medium–High	5 hours
Statistical Software Practice: Hands-on with Python/R/Excel for descriptive and inferential stats	Easy–Medium	5 hours
Field Study/Survey: Design and conduct a small survey, then analyze the data (mean, SD, correlation)	Medium	5 hours

H. Learner centric activities:

Tutorial sessions include problem-solving on statistical measures, probability, and data interpretation through peer discussions and hands-on tools. Activities aim to enhance conceptual clarity and analytical thinking.

I. Course Articulation Matrix: (Mapping of COs with POs)

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES			CORRELATION WITH PROGRAM SPECIFIC OUTCOMES	
		PO1	PO2	PO3	PSO1	PSO2
[AI6101.1]	Explain the foundational principles of statistics and their applications in engineering contexts.	2	1	2	2	1
[AI6101.2]	Apply appropriate measures of central tendency, dispersion, skewness, and kurtosis to analyses data sets.	2	2	2	2	2
[AI6101.3]	Analyze various probability distributions and use them to model random processes.	3	2	3	3	3
[AI6101.4]	Evaluate relationships between variables using correlation and regression techniques.	3	2	3	3	2
[AI6101.5]	Perform hypothesis testing using appropriate statistical tests to draw inferences from data.	3	3	3	3	2

1-Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation

Course Coordinator
Dr. Rahul Sharma

Dr. Deepak Panwar
Head of Dept.
CSE (AIML)

Student Representative
Name:
Registration No.:

